Quantitative Comparisons

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Numeric comparisons**

Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$A = 56$. $B = 54 + 2 = 56$. $A = B$.

2 **[Foundational]** • **Target: Numeric comparisons**

Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$A = 579$. $B = 580$. $B > A$.

3 **[Foundational]** • **Target: Numeric comparisons**

Column A: 9^2. Column B: $8^2 + 15$. Compare the two columns.

$A = 81$. $B = 64 + 15 = 79$. $A > B$.

4 **[Foundational]** • **Target: Numeric comparisons**

Column A: $(-3)^2$. Column B: -3^2. Compare the two columns.

$A = (-3) \times (-3) = 9$. $B = -(3^2) = -9$ (no parentheses around the base). $A > B$.

5 **[Foundational]** • **Target: Fraction and decimal comparisons**

Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

$\frac{3}{7} \approx 0.4286$, which is less than 0.43. $B > A$.

6 **[Foundational]** • **Target: Fraction and decimal comparisons**

Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.

$2\frac{3}{8} = \frac{19}{8}$. $A = B$.

7 **[Foundational]** • **Target: Fraction and decimal comparisons**

Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

$\frac{5}{9} \approx 0.5556$, which is less than 0.6. $A > B$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Percent comparisons

Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$A = 0.25 \times 160 = 40$. $B = 0.20 \times 200 = 40$. $A = B$.

9 [Intermediate] • Target: Percent comparisons

Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$A = 100 \times 1.1 \times 1.1 = 121$. $B = 100 \times 1.2 = 120$. $A > B$.

10 [Intermediate] • Target: Exponent and root comparisons

Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.

$\sqrt{50} \approx 7.07$, which is less than 7.1. $B > A$.

11 [Intermediate] • Target: Exponent and root comparisons

Column A: 2^{10}. Column B: 10^3. Compare the two columns.

$A = 1{,}024$. $B = 1{,}000$. $A > B$.

12 [Intermediate] • Target: Exponent and root comparisons

Column A: 3^4. Column B: 4^3. Compare the two columns.

$A = 81$. $B = 64$. $A > B$.

13 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

$A = 2(5) + 3 = 13$. $B = 5 + 8 = 13$. $A = B$.

14 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

$(x + 2)^2$ expands to $x^2 + 4x + 4$ for every value of x, so the columns are always equal. $A = B$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 6.01 Sequences & Patterns

Given $n = -2$. Column A: n^2. Column B: n^3. Compare the two columns.

$A = (-2)^2 = 4$. $B = (-2)^3 = -8$. $A > B$.

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16 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 7.01 Data Analysis & Probability
Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.

$A = 5(3) - 2 = 13$. $B = 2(3 + 5) = 16$. $B > A$.

17 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 8.01 Word Problems & Reasoning
For any real number x: Column A: x. Column B: x^2. Compare the two columns.

If $x = 2$: $B > A$. If $x = \frac{1}{2}$: $A > B$. If $x = 1$: $A = B$. The relationship changes, so it cannot be determined.

18 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 1.01 Arithmetic & Number Sense
For any real number x: Column A: $x + 5$. Column B: $2x$. Compare the two columns.

If $x = 5$: $A = B$. If $x = 0$: $A > B$. If $x = 10$: $B > A$. The relationship changes, so it cannot be determined.

19 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 2.01 Ratios, Proportions & Percentages
For any real number x: Column A: $|x|$. Column B: x. Compare the two columns.

If $x \geq 0$: $A = B$. If $x < 0$ (e.g. $x = -3$): $A = 3$, $B = -3$, so $A > B$. Since the relationship is not the same in every case, it cannot be determined.

20 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 3.01 Algebra Essentials
Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

If $n = 1$: $A = 1 < 4$. If $n = 3$: $A = 9 > 4$. The relationship changes, so it cannot be determined.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Geometry comparisons
Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

$C = 2(3.14)(5) = 31.4$, which is less than 32.

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22 [Foundational] • Target: Geometry comparisons

Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. **Column A is greater ✓**

Rectangle area $= 9 \times 4 = 36$; square perimeter $= 4 \times 7 = 28$. Column A is greater.

23 [Foundational] • Target: Geometry comparisons

Column A: the hypotenuse of the right triangle shown. Column B: 16.

a. The relationship cannot be determined from the given information

b. **Column B is greater ✓**

c. The two columns are equal

d. Column A is greater

$\sqrt{9^2 + 12^2} = \sqrt{225} = 15$, which is less than 16.

24 [Foundational] • Target: More geometry comparisons

Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. **Column B is greater ✓**

d. The two columns are equal

Cube $= 5^3 = 125$; box $= 4 \times 5 \times 7 = 140$. Column B is greater.

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25 [Foundational] • Target: More geometry comparisons

Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.

a. Column B is greater

b. The two columns are equal

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Octagon $= (8-2) \times 180 = 1{,}080^\circ$; four triangles $= 4 \times 180 = 720^\circ$. Column A is greater.

26 [Foundational] • Target: Numeric comparisons

Column A: 6×7. Column B: $5 \times 9 - 3$. Compare the two columns.

a. **The two columns are equal ✓**

b. Column B is greater

c. The relationship cannot be determined from the given information

d. Column A is greater

42 vs $45 - 3 = 42$: equal.

27 [Foundational] • Target: Numeric comparisons

Column A: $345 + 678$. Column B: $900 + 100$. Compare the two columns.

a. **Column A is greater ✓**

b. The relationship cannot be determined from the given information

c. Column B is greater

d. The two columns are equal

$1{,}023$ vs $1{,}000$: Column A is greater.

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28 [Foundational] • Target: Percent comparisons

Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.

a. The relationship cannot be determined from the given information

b. **The two columns are equal ✓**

c. Column A is greater

d. Column B is greater

$100 \times 0.8 \times 1.2 = 96$: equal.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 6$. Column A: $3x - 4$. Column B: $x + 10$. Compare the two columns.

a. **Column B is greater ✓**

b. Column A is greater

c. The relationship cannot be determined from the given information

d. The two columns are equal

14 vs 16: Column B is greater.

30 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 2$. Column A: $4x + 1$. Column B: $3(x + 2)$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

9 vs 12: Column B is greater.

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31 [Intermediate] • Target: Comparisons requiring testing values

For any real number x: Column A: x^2. Column B: 0. Compare the two columns.

a. The relationship cannot be determined from the given information ✓

b. The two columns are equal

c. Column A is greater

d. Column B is greater

x^2 is always ≥ 0, but could equal 0 (at $x = 0$) or exceed it — not always the same relationship.

32 [Intermediate] • Target: Comparisons requiring testing values

For any integer n: Column A: n. Column B: $-n$. Compare the two columns.

a. The two columns are equal

b. The relationship cannot be determined from the given information ✓

c. Column B is greater

d. Column A is greater

If $n > 0$, A is greater; if $n < 0$, B is greater; if $n = 0$, they are equal — it varies.

33 [Intermediate] • Target: Comparisons requiring testing values

For any real number x with $x \neq 0$: Column A: x^2. Column B: 0. Compare the two columns.

a. Column B is greater

b. Column A is greater ✓

c. The two columns are equal

d. The relationship cannot be determined from the given information

Since $x \neq 0$, x^2 is always strictly positive, so Column A is always greater.

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34 [Intermediate] • Target: Comparisons requiring testing values

For any real number x: Column A: $(x+2)^2$. Column B: x^2+4x+4. Compare the two columns.

a. The two columns are equal ✓

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

$(x+2)^2 = x^2+4x+4$ for every x: equal.

35 [Intermediate] • Target: Comparisons requiring testing values

Given only that m is a positive integer: Column A: m. Column B: m^2. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information ✓

At $m = 1$ they are equal; for $m > 1$, Column B is greater — it varies.

36 [Intermediate] • Target: Comparisons requiring testing values

For any real number x: Column A: $-x$. Column B: $x-1$. Compare the two columns.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information ✓

d. Column B is greater

$-x-(x-1) = 1-2x$, whose sign depends on x — it varies.

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37 [Intermediate] • Target: Geometry comparisons

Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. **Column A is greater ✓**

d. Column B is greater

Hexagon $= 720^\circ$; two triangles $= 180 + 180 = 360^\circ$. Column A is greater.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometry comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

$V = 3.14 \times 9 \times 7 = 197.82$, which is less than 200.

39 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Column A: the value of x, if $x : 15 = 2 : 5$. Column B: 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. **Column B is greater ✓**

$x = 15 \times \frac{2}{5} = 6$, which is less than 7.

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40 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 3.01 Algebra Essentials

Two numbers are in the ratio $4:7$, and their sum is 99. Column A: the smaller number. Column B: 40.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

Each part $= 99 \div 11 = 9$; smaller $= 4 \times 9 = 36$, which is less than 40.

41 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 4.01 Geometry & Measurement

Column A: the sum of the parts in the simplified form of the ratio $24:36$. Column B: 5.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **The two columns are equal ✓**

$24:36$ simplifies to $2:3$; $2 + 3 = 5$: equal.

42 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 6.01 Sequences & Patterns

A recipe uses water and rice in the ratio $2:1$. Column A: the water needed for 300 g of rice. Column B: 550 g.

a. Column B is greater

b. **Column A is greater ✓**

c. The two columns are equal

d. The relationship cannot be determined from the given information

Water $= 2 \times 300 = 600$ g, which is greater than 550 g.

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43 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Mean $= \frac{45}{5} = 9$; median $= 8$. Column A is greater.

44 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the range of the data set $6, 11, 19, 25$. Column B: 19.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. **The two columns are equal ✓**

d. Column A is greater

Range $= 25 - 6 = 19$: equal.

45 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

a. **Column B is greater ✓**

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column A is greater

$P = \frac{4}{13} \approx 0.31$, which is less than 0.4: Column B is greater.

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